CASE STUDY

Smart Food Delivery Management System Using Data Structures and Algorithms

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Problem Statement-

Food delivery platforms receive multiple orders from different customers and restaurants at the same time. Efficiently managing these orders, prioritizing urgent deliveries, assigning available delivery partners, and finding suitable routes between restaurants and customers are important for timely delivery.

The proposed system aims to develop a **Smart Food Delivery Management System using Data Structures and Algorithms**. The system manages incoming orders using queues and priority queues, maintains delivery partner information using linked lists, and represents the delivery network using a weighted graph. Dijkstra's algorithm is used to determine the shortest route between the restaurant and customer.

The project demonstrates how fundamental Data Structures and Algorithms can be applied to solve practical problems in a food delivery environment.

1. Introduction

Food delivery platforms receive a large number of orders from customers throughout the day. During peak hours, such as lunch and dinner, several orders may arrive at the same time. These orders may have different priorities and may originate from different restaurants and locations.

At the same time, delivery partners are distributed across different locations and may have different availability statuses. After an order is processed, an efficient route must be identified between the restaurant and the customer's location.

This case study demonstrates how **Data Structures and Algorithms** can be applied to manage these challenges in a food delivery environment.

2. Case Study Scenario

Consider a food delivery service operating across a network of connected locations.

The system receives orders from customers who order food from different restaurants. Each order contains information such as:

- Order ID
- Customer ID
- Restaurant ID
- Restaurant location
- Customer location
- Food item
- Quantity
- Priority

The system also maintains information about delivery partners, including their:

- Partner ID
- Name
- Current location
- Availability status

During peak hours, orders cannot simply be processed randomly. Urgent orders need to be given higher priority, suitable delivery partners need to be assigned, and an efficient route needs to be determined for the delivery.

3. Problem Identified

The food delivery system faces three major problems.

Problem 1 — Managing Multiple Orders

Orders arrive continuously and need to be stored and processed systematically.

If all orders are processed randomly, the system may fail to handle urgent orders efficiently.

Problem 2 — Delivery Partner Assignment

Several delivery partners may be available at different locations. The system needs to identify an available partner who can reach the restaurant efficiently.

Problem 3 — Route Optimization

After a delivery partner is assigned, the system needs to determine an efficient route from the restaurant to the customer's location.

4. Proposed Solution

Problem	**Solution**
Managing incoming orders	Queue / Priority Queue
Processing urgent orders	Priority Queue
Maintaining delivery partners	Linked List
Maintaining order history	Linked List
Representing roads	Weighted Graph
Finding shortest route	Dijkstra's Algorithm

5. ALGORITHMS AND PSEUDOCODE

Algorithm: Add Order

Purpose: To add a new food order to the priority queue.

Algorithm

1. Start.
2. Read order ID, customer ID, restaurant ID, restaurant location, customer location, food item, quantity and priority.
3. Check whether the priority queue is full.
4. If full, display “Queue is full”.
5. Otherwise, insert the order into the priority queue.
6. Increase the priority queue size.
7. Display “Order added successfully”.
8. Stop.

Algorithm: Find Highest Priority Order

Purpose: To find the order that should be processed next.

Algorithm

1. Start.
2. Check whether there are any pending orders.
3. If there are no orders, display “No pending orders”.
4. Otherwise, compare the priority of all pending orders.
5. Select the order with the smallest priority value.
6. Return the selected order.
7. Stop.

Priority values:

- 1 → Urgent

- 2 → High
- 3 → Normal

Algorithm: Add Delivery Partner

Purpose: To maintain delivery partner information using a linked list.

Algorithm

1. Start.
2. Read partner ID, name and current location.
3. Set partner status as available.
4. Create a new linked-list node.
5. Add the node to the delivery partner list.
6. Display confirmation.
7. Stop.

Algorithm: Assign Delivery Partner

Purpose: To assign the nearest available delivery partner to an order.

Algorithm

1. Start.
2. Select the order to be processed.
3. Search the linked list of delivery partners.
4. Consider only available partners.
5. Calculate the distance from each available partner to the restaurant using Dijkstra's algorithm.
6. Select the partner with the minimum distance.
7. Mark the selected partner as unavailable.
8. Display the assigned partner.

9. Stop.

Algorithm: Add Road

Purpose: To represent the delivery road network using a weighted graph.

Algorithm

1. Start.
2. Read the source location, destination location and distance.
3. Store the distance in the adjacency matrix.
4. Since the road is bidirectional, store the distance in both directions.
5. Stop.

Algorithm: Dijkstra's Shortest Path

Purpose: To find the shortest route between two locations.

Algorithm

1. Start.
2. Select the source location.
3. Set the distance of the source to 0.
4. Set the distance of all other locations to infinity.
5. Mark all locations as unvisited.
6. Select the unvisited location having the smallest distance.
7. Mark it as visited.
8. Check all neighbouring locations.
9. Calculate the new distance through the selected location.
10. If the new distance is smaller, update the distance and parent.
11. Repeat until all required locations are processed.
12. Display the shortest distance and route.

13. Stop.

Algorithm: Process Order

Purpose: To process the next order, assign a delivery partner and find the delivery route.

Algorithm

1. Start.
2. Find the highest-priority pending order.
3. If no order exists, display a message.
4. Find the nearest available delivery partner.
5. If no partner is available, keep the order pending.
6. Find the shortest route from the restaurant to the customer using Dijkstra's algorithm.
7. Mark the order as delivered.
8. Add the completed order to the order-history linked list.
9. Remove the order from the priority queue.
10. Display the delivery details.
11. Stop.

Algorithm: Cancel Order

Purpose: To cancel a pending order and store it in order history.

Algorithm

1. Start.
2. Read the order ID.
3. Search for the order in the priority queue.
4. If the order is not found, display "Order not found".
5. Otherwise, change its status to cancelled.
6. Add the order to the history linked list.

7. Remove the order from the priority queue.
8. Display confirmation.
9. Stop.

Algorithm: Add Order to History

Purpose: To maintain completed and cancelled orders using a linked list.

Pseudocode

START

Create new history node

Store:

Order ID

Customer ID

Food item

Order status

IF history list is empty

Make new node the first node

ELSE

Traverse to the end

Add new node

END IF

STOP

Overall System Algorithm

START

Initialize priority queue

Initialize delivery partner linked list

Initialize order history linked list

Initialize road network

REPEAT

Display main menu

Read user's choice

CASE choice

1:

Add Order

2:

Display Pending Orders

3:

Process Next Order

4:

Add Delivery Partner

5:

Display Delivery Partners

6:

Assign Delivery Partner

7:

Add Road

8:

Display Road Network

9:

Find Shortest Delivery Route

10:

Display Order History

11:

Cancel Order

12:

Exit program

OTHERWISE:

Display "Invalid choice"

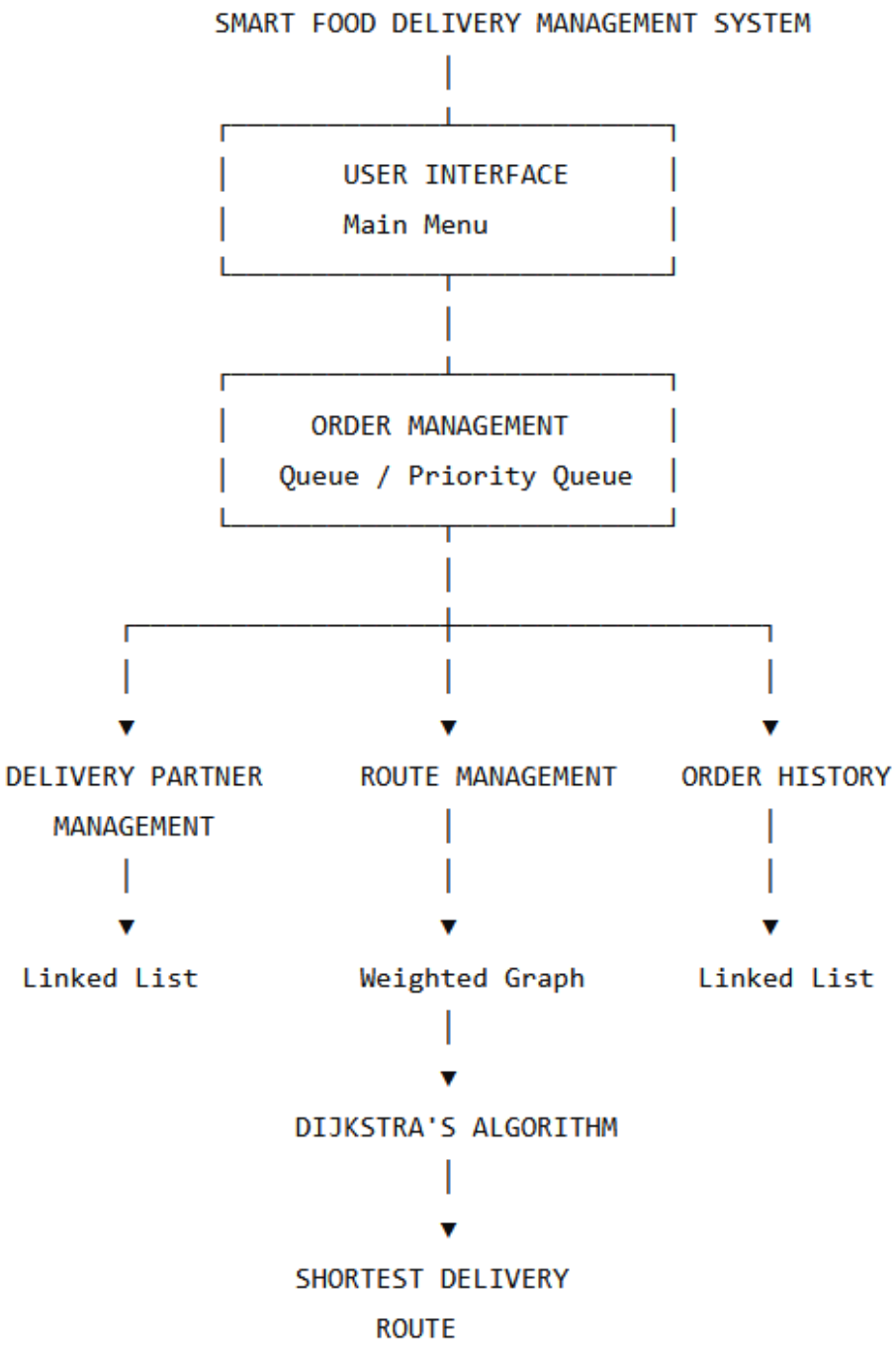
END CASE

UNTIL choice = 12

STOP

Main DSA Used

Problem	**Data Structure / Algorithm**
Managing pending orders	Priority Queue
Prioritizing urgent orders	Priority Queue
Delivery partner records	Linked List
Order history	Linked List
Road network	Weighted Graph
Shortest delivery route	Dijkstra's Algorithm
Partner selection	Linked List + Dijkstra


SMART FOOD DELIVERY MANAGEMENT SYSTEM
USER INTERFACE
Main Menu
ORDER MANAGEMENT
Queue / Priority Queue
DELIVERY PARTNER MANAGEMENT
ROUTE MANAGEMENT
ORDER HISTORY
Linked List
Weighted Graph
Linked List
DIJKSTRA'S ALGORITHM
SHORTEST DELIVERY ROUTE

OUTPUTS:-

Adding delivery partners

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 4

Enter Partner ID: 201
Enter Partner Name: Mukesh
Enter Current Location: 5

Delivery partner added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 4

Enter Partner ID: 202
Enter Partner Name: Praveen
Enter Current Location: 3

Delivery partner added successfully.
```

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 4

Enter Partner ID: 203
Enter Partner Name: Gaurav
Enter Current Location: 7

Delivery partner added successfully.
```

Adding roads:-

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 1
Enter Destination Location: 2
Enter Distance: 4

Road added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 1
Enter Destination Location: 3
Enter Distance: 3

Road added successfully.
```

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 2
Enter Destination Location: 4
Enter Distance: 2

Road added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 2
Enter Destination Location: 5
Enter Distance: 5

Road added successfully.
```

```
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 4
Enter Destination Location: 6
Enter Distance: 3

Road added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 5
Enter Destination Location: 6
Enter Distance: 3

Road added successfully.
```

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 5
Enter Destination Location: 7
Enter Distance: 3

Road added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 7

Enter Source Location: 66
Enter Destination Location: 7
Enter Distance: 2

Invalid location.
```

Adding orders:-

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 1

Enter Order ID: 501
Enter Customer ID: 101
Enter Restaurant ID: 1
Enter Restaurant Location: 2
Enter Customer Location: 7
Enter Food Item: pizza
Enter Quantity: 2

Priority:
1. Urgent
2. High
3. Normal
Enter Priority: 2

Order added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 1

Enter Order ID: 502
Enter Customer ID: 102
Enter Restaurant ID: 2
```

```
Enter Order ID: 502
Enter Customer ID: 102
Enter Restaurant ID: 2
Enter Restaurant Location: 4
Enter Customer Location: 6
Enter Food Item: burger
Enter Quantity: 1

Priority:
1. Urgent
2. High
3. Normal
Enter Priority: 3

Order added successfully.


==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 1

Enter Order ID: 503
Enter Customer ID: 103
Enter Restaurant ID: 3
Enter Restaurant Location: 5
Enter Customer Location: 3
Enter Food Item: noodles
Enter Quantity: 1

Priority:
1. Urgent
2. High
3. Normal
Enter Priority: 1

Order added successfully.
```

Display delivery partners:-

```
========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 5

========== DELIVERY PARTNERS ==========

Partner ID: 201
Name: Mukesh
Location: 5
Status: Available
--------------------------------------
Partner ID: 202
Name: Praveen
Location: 3
Status: Available
--------------------------------------
Partner ID: 203
Name: Gaurav
Location: 7
Status: Available
--------------------------------------
```

Display road networls:-

```
========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 8

========== ROAD NETWORK ==========

Location 0:
Location 1: -> Location 2 (4 km) -> Location 3 (3 km)
Location 2: -> Location 1 (4 km) -> Location 4 (2 km) -> Location 5 (5 km)
Location 3: -> Location 1 (3 km) -> Location 6 (4 km)
Location 4: -> Location 2 (2 km) -> Location 6 (3 km)
Location 5: -> Location 2 (5 km) -> Location 6 (3 km) -> Location 7 (3 km)
Location 6: -> Location 3 (4 km) -> Location 4 (3 km) -> Location 5 (3 km)
Location 7: -> Location 5 (3 km)
```

Process next order:-

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 3

========== PROCESSING ORDER ==========
Order ID: 503
Food: noodles

Delivery Partner Assigned
Partner ID: 201
Partner Name: Mukesh
Distance from Restaurant: 0 km

Shortest Delivery Route: 5 -> 6 -> 3
Total Delivery Distance: 7 km

Order 503 delivered successfully.
```

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 2

========== PENDING ORDERS ==========

Order ID: 501
Customer ID: 101
Restaurant ID: 1
Food: pizza
Quantity: 2
Priority: High
----------------------------------
Order ID: 502
Customer ID: 102
Restaurant ID: 2
Food: burger
Quantity: 1
Priority: Normal
----------------------------------
```

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 3

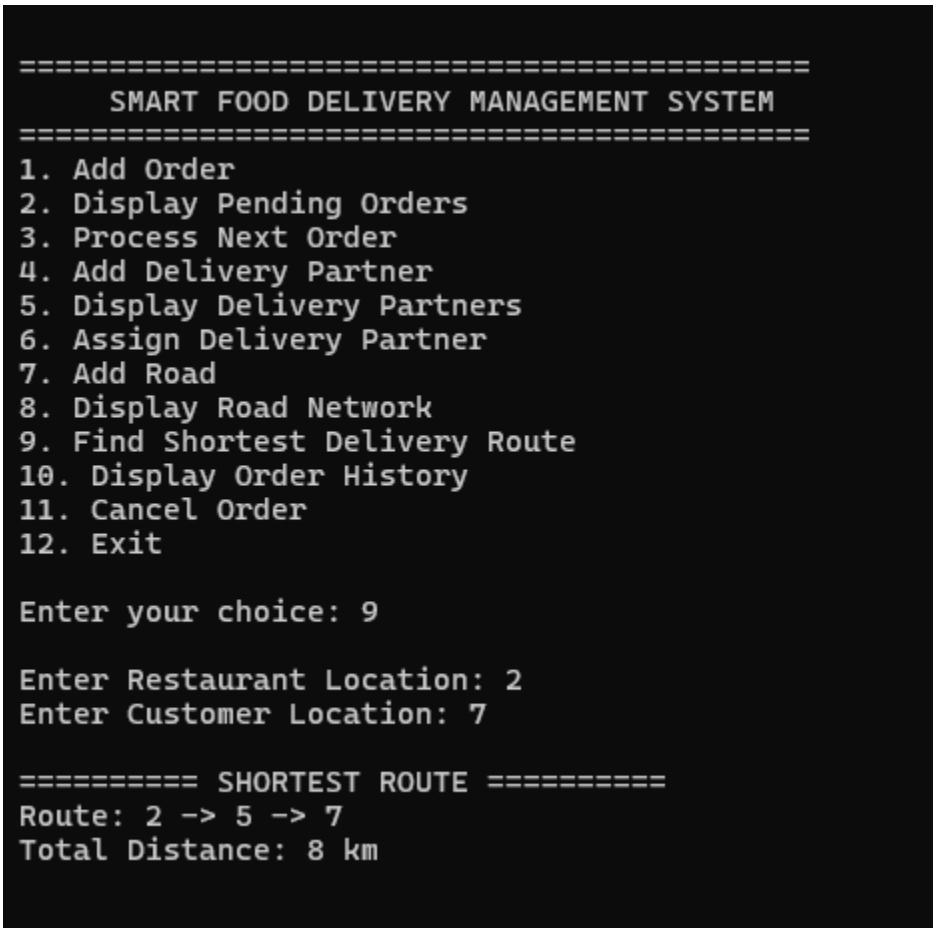
========== PROCESSING ORDER ==========
Order ID: 501
Food: pizza

Delivery Partner Assigned
Partner ID: 202
Partner Name: Praveen
Distance from Restaurant: 7 km

Shortest Delivery Route: 2 -> 5 -> 7
Total Delivery Distance: 8 km

Order 501 delivered successfully.
```

Finding shortest delivery route:-



Canceling order:-

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 11

Enter Order ID to cancel: 502

Order 502 cancelled successfully.
```

EXIT :-

```
==========================================
     SMART FOOD DELIVERY MANAGEMENT SYSTEM
==========================================
1. Add Order
2. Display Pending Orders
3. Process Next Order
4. Add Delivery Partner
5. Display Delivery Partners
6. Assign Delivery Partner
7. Add Road
8. Display Road Network
9. Find Shortest Delivery Route
10. Display Order History
11. Cancel Order
12. Exit

Enter your choice: 12

Thank you for using the system.

Process returned 0 (0x0)   execution time : 429.681 s
Press any key to continue.
```

12. DSA Application

The case study demonstrates the following:

Priority Queue

Used to determine which order should be processed first.

Urgent → High → Normal

Linked List

Used to dynamically maintain delivery partners and order history.

Partner 201 → Partner 202 → Partner 203 → NULL

Graph

Used to represent the road network.

Location = Vertex

Road = Edge

Distance = Weight

Dijkstra's Algorithm

Used to find the shortest route between two locations.

13. Results

The system successfully demonstrates a complete food delivery workflow:

Order Placement

↓

Priority-Based Processing

↓

Delivery Partner Assignment

↓

Route Calculation

↓

Food Delivery

↓

Order History

Order	Priority	Route	Distance	Result
503	Urgent	5 → 6 → 3	7 km	Delivered
501	High	2 → 5 → 7	8 km	Delivered
502	Normal	—	—	Cancelled

14. Complexity Analysis

Operation	Complexity
Add order	O(1)
Find highest-priority order	O(n)
Remove processed order	O(n)
Add delivery partner	O(n)
Search available partner	O(n)
Dijkstra using adjacency matrix	$O(V^2)$
Add order-history node	O(n)

Where:

- **n** = number of orders/partners
- **V** = number of locations

15. Conclusion

The case study demonstrates how Data Structures and Algorithms can be applied to a real-world food delivery management problem.

The system uses a **priority-based order management mechanism** to handle orders with different priorities, linked lists to maintain dynamic records, and a weighted graph to represent the delivery network. **Dijkstra's Algorithm** is used to determine shortest routes between restaurants and customers.

The project therefore combines multiple fundamental DSA concepts into a single practical application and demonstrates their role in managing the food delivery process efficiently.